

8	A <u>Option A:</u> WD to change speed of object from 2 m s^{-1} to $3 \text{ m s}^{-1} = \frac{1}{2} m (3^2 - 2^2)$ $= \frac{1}{2} m (5)$ WD to change speed of object from 1 m s^{-1} to $2 \text{ m s}^{-1} = \frac{1}{2} m (2^2 - 1^2)$ $= \frac{1}{2} m (3)$ So option A is correct. <u>Option B:</u> When an object slows down, there has to be negative work done by a force (opposing the motion of the object). Hence option B is not correct. <u>Option C:</u> When lifting an object 1 m at constant velocity, $WD = mgh = mg(1)$ When moving an object 1 m across a smooth horizontal surface at constant speed, there is no need for a force to keep it at constant velocity. Therefore, WD is zero. Hence option C is not correct. <u>Option D:</u> When an object is dropped, the force of gravity is in the direction of the motion of the object. There is WD by the force of gravity and is given by $W \times d$. Hence option D is not correct.	A
9	C	D